

Consider the matrices below with $X = AB$, where

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 3 & -1 & 1 & 3 \\ -2 & 5 & -k & k+1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 1 & 2 & & -1 \\ 0 & 4 & & 1 \\ 3 & -1 & & 1 \\ -2 & 5 & k(k-1) & \end{bmatrix}$$

For which value(s) of k will the entry X_{23} be zero?

- ☐ $k = 3$
- ☐ No such k
- ☐ $k = -2$
- ☒ $k = \frac{1 \pm \sqrt{5}}{2}$
- ☐ $k = 0$ or $k = 1$
- ☐ None of the given option

Clear my choice